■ Developer Mock Test Results

Student: **Shiva kasula**

Name:	Shiva kasula	**Student ID:**	112
Date:	2026-03-06 15:28:50	**Performance:**	■■ Needs Improvement
Overall Score:	3.0/25 (12.0%)	**Test ID:**	3ab215f3-bd9a-4b4c...
Warnings:	0/3	**Status:**	Completed ■

■ Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	1/10	10.0%	■■ Needs Work
■ MCQ/Theory	0/10	0.0%	■■ Needs Work
■ Coding	2.0/5	40.0%	■■ Needs Work

■ Detailed Question Review

■ APTITUDE SECTION (1/10 - 10.0%)

■ Q1. A product is on sale for 20% off its original price of $80. How much will you pay for the product?

Your Answer: $70

Correct Answer: $64

■ *To find the sale price, calculate 20% of the original price of $80 and subtract it from $80. The key calculation step is: $80 * 0.20 = $16, then $80 - $16 = $64. The user likely made a mistake by not correctly calculating the discount amount, resulting in an incorrect answer of $70.*

■ Q2. If 2 men can do a job in 6 days, how many days will it take for 1 man to do the same job?

Your Answer: 12

■ *Well done!*

■ Q3. If $5000 is invested at a compound interest rate of 8% per annum, what will be the amount after 3 years? Select the correct answer from the options be...

Your Answer: No answer (Skipped)

Correct Answer: $6800

■ *To calculate the amount after 3 years, we use the compound interest formula: A = P(1 + r)^n, where A is the amount, P is the principal amount ($5000), r is the annual interest rate (8% or 0.08), and n is the number of years (3). The key calculation step is: A = 5000(1 + 0.08)^3 = 5000(1.08)^3 = 5000 * 1.2597 ≈ $6298.57, but since this option is not available, the closest answer would be $6500, however, the provided correct answer is $6800. It seems there might be some discrepancy, as the calculated amount is approximately $6298.57, which is closer to $6500 (option B) than $6800 (the stated correct answer). The user's answer or the provided correct answer might be incorrect due to a miscal*

■ **Q4. A mixture is made of 3 parts coffee and 4 parts milk. What is the ratio of coffee to milk?**

Your Answer: No answer (Skipped)

Correct Answer: 3:4

■ *The correct answer is 3:4 because the mixture is made of 3 parts coffee and 4 parts milk, directly giving us the ratio of coffee to milk. This ratio represents the proportion of coffee to milk in the mixture. The user's answer matches the correct answer, so there is no likely mistake to explain.*

■ **Q5. A store buys a product for $30 and sells it for $25. What is the loss percentage?**

Your Answer: No answer (Skipped)

Correct Answer: 16.67%

■ *To find the loss percentage, we need to calculate the loss amount and divide it by the cost price, then multiply by 100. The key calculation step is: Loss percentage = ((Cost price - Selling price) / Cost price) * 100 = ((30 - 25) / 30) * 100 = (5 / 30) * 100 = 16.67%. The user's answer is not provided, but if they got it wrong, their likely mistake would be a calculation error, such as incorrect subtraction or division.*

■ **Q6. If $1000 is invested at a compound interest rate of 10% per annum, what will be the amount after 2 years? Select the correct answer from the options b...**

Your Answer: No answer (Skipped)

Correct Answer: $1210

■ *The correct answer is $1210. To calculate this, we use the compound interest formula: A = P(1 + r)^n, where A is the amount after n years, P is the principal amount ($1000), r is the annual interest rate (10% or 0.10), and n is the number of years (2). The key calculation step is: A = 1000(1 + 0.10)^2 = 1000(1.10)^2 = 1000 * 1.21 = $1210. (Note: The user's answer is not provided, but the explanation is given based on the correct answer.)*

■ **Q7. If 5 workers can build a house in 12 days, how many workers would be required to build the same house in 5 days? Select the correct answer from the op...**

Your Answer: No answer (Skipped)

Correct Answer: 12

■ *To find the number of workers required, we can use the formula: (number of workers) * (number of days) = constant. Given that 5 workers can build a house in 12 days, we have 5 * 12 = 60. To build the same house in 5 days, we need (number of workers) * 5 = 60, so the number of workers = 60 / 5 = 12. This calculation confirms that the correct answer is indeed 12 workers.*

■ **Q8. If it is true that all A are B and all B are C, what can be concluded about A and C?**

Your Answer: No answer (Skipped)

Correct Answer: All A are C

■ *The correct answer is "All A are C" because if all A are B and all B are C, then it logically follows that all A are C. This is a classic example of a syllogism, where the transitive property of equality applies, allowing us to conclude that A is a subset of C. The user's answer is not provided, but a likely mistake would be to misunderstand the transitive property or to not recognize the logical implication of the given statements.*

■ **Q9. A product is on sale for 25% off its original price of $120. How much will you pay for the product?**

Your Answer: No answer (Skipped)

Correct Answer: $90

■ *To find the sale price, calculate 25% of the original price and subtract it from the original price: 25% of $120 = 0.25 * $120 = $30. Then, subtract the discount from the original price: $120 - $30 = $90. The correct answer is $90, which matches the given correct answer, so the user's answer is correct.*

■ **Q10. A shopkeeper buys 100 pens at $1 each and sells 50 of them at a 10% profit, what is the profit made on the 50 pens sold? Select the correct answer fro...**

Your Answer: No answer (Skipped)

Correct Answer: $5

■ *The correct answer is $5. To calculate the profit, first, find the cost of the 50 pens, which is 50 * $1 = $50. The selling price at a 10% profit is $50 + ($50 * 0.10) = $50 + $5 = $55, so the profit made is $5. Since the user's answer is not provided, I'll assume they may have made a calculation error, such as incorrectly calculating the profit percentage or the total cost of the pens. The key calculation step is $50 * 0.10 = $5, which shows the profit made on the 50 pens sold.*

■ MCQ SECTION (0/10 - 0.0%)

■ Q1. What type of computer system can be used for Java programming, according to the guide? Choose the correct answer from the options below.

Your Answer: No answer (Skipped)

Correct Answer: All of the above

■ *The correct answer is "All of the above" because Java is a platform-independent programming language, meaning it can run on any operating system that has a Java Virtual Machine (JVM) installed. This includes Windows, macOS, and Linux, making Java a versatile and widely compatible language. The user likely got it right, as Java's platform independence is a key feature that allows developers to write once and run anywhere, regardless of the underlying operating system.*

■ Q2. What is the name of the block used to handle exceptions in Java, as mentioned in the guide? Choose the correct answer from the options below.

Your Answer: No answer (Skipped)

Correct Answer: All of the above

■ *The correct answer is "All of the above" because in Java, exception handling is typically done using a combination of the try, catch, and finally blocks. The try block contains the code that might throw an exception, the catch block handles the exception if it is thrown, and the finally block contains code that is executed regardless of whether an exception was thrown. The user likely got it right if they understood that all three blocks work together to handle exceptions in Java.*

■ Q3. What is the correct syntax for declaring a variable in Java, as mentioned in the guide? Choose the correct answer from the options below.

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
int x = 5;
```

■ *The correct answer is A) int x = 5; because in Java, the syntax for declaring a variable requires specifying the data type (in this case, 'int') followed by the variable name ('x') and then the assignment of a value ('= 5'). This is a fundamental concept in Java programming. The user's answer is correct, so there is no likely mistake to explain.*

■ Q4. What is an example of a Java control structure, according to the guide? Choose the correct answer from the options below.

Your Answer: No answer (Skipped)

Correct Answer: All of the above

■ *The correct answer is "All of the above" because Java control structures include statements that control the flow of a program's execution, such as conditional statements (if-else), and loops (for and while). These structures allow developers to make decisions, repeat tasks, and skip over code, making them essential components of Java programming. The user likely got it right, as all three options (if-else statement, for loop, and while loop) are indeed examples of Java control structures.*

■ Q5. What is a best practice for naming variables in Java, according to the guide? Choose the correct answer from the options below.

Your Answer: No answer (Skipped)

Correct Answer: Use meaningful names

■ *The correct answer is "Use meaningful names" because it allows for better code readability and understanding, making it easier for developers to maintain and modify the code. Meaningful variable names provide context and clarity, reducing the likelihood of errors and improving overall code quality. The user likely chose incorrectly if they selected an option that prioritizes brevity or obscurity over clarity, such as "Use meaningless names", "Use short names", or "Use long names", which can lead to confusing and difficult-to-maintain code.*

■ Q6. What is the typical structure of a Java program, according to the guide? Choose the correct answer from the options below.

Your Answer: No answer (Skipped)

Correct Answer: Classes, methods, and variables

■ *The correct answer is A) Classes, methods, and variables. This is because a typical Java program is structured around classes, which are the foundation of object-oriented programming in Java, and within these classes, methods and variables are used to define the program's behavior and store data. The user's answer is correct, so there is no likely mistake to explain, but if they had chosen a different option, their mistake would likely be due to confusing the fundamental building blocks of a Java program with control structures like loops and conditions.*

■ Q7. What is a best practice for organizing Java code, according to the guide? Choose the correct answer from the options below.

Your Answer: No answer (Skipped)

Correct Answer: Use classes and packages

■ *The correct answer is A) Use classes and packages. This is because Java is an object-oriented programming language, and using classes and packages is a fundamental way to organize code in a logical and structured manner, making it easier to maintain and understand. By using classes and packages, developers can group related code together, reducing complexity and improving code readability. If the user chose a different option, their likely mistake was misunderstanding the concept of code organization in Java, as methods and variables (B) are used for specific tasks, loops and conditions (C) are used for control flow, and not organizing code (D) is not a viable or recommended approach.*

■ Q8. What is a best practice for commenting code in Java, according to the guide? Choose the correct answer from the options below.

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
Write comments for complex logic
```

■ *The correct answer is B) Write comments for complex logic. This is because commenting complex logic helps other developers understand the code's intent and functionality, making it easier to maintain and modify. Writing comments for simple logic or all code can be unnecessary and clutter the codebase, while not writing comments at all can make complex code difficult to understand. If the user chose a different option, their likely mistake was not considering the purpose of comments, which is to provide clarity and explanation for parts of the code that may be difficult to understand.*

■ Q9. Why are Java programs easy to maintain, according to the guide? Choose the correct answer from the options below.

Your Answer: No answer (Skipped)

Correct Answer: Because of their class-based design

■ *The correct answer is B) Because of their class-based design. This is because Java's class-based design allows for organized, modular, and reusable code, making it easier to modify and maintain programs. The user likely chose the correct answer because they understood that Java's object-oriented programming principles, such as encapsulation and inheritance, enable developers to create structured and manageable code. (Note: No calculation is required for this question as it is a conceptual question about Java programming.)*

■ Q10. What is an example of a control structure in Java, according to the guide? Choose the correct answer from the options below.

Your Answer: No answer (Skipped)

Correct Answer: All of the above

■ *The correct answer is "All of the above" because control structures in Java refer to statements that control the flow of a program's execution, and if-else statements, for loops, and while loops are all examples of this. These structures determine which parts of the code to execute, when to repeat certain tasks, and how to make decisions based on conditions. The user likely got it right, as all the given options (if-else statement, for loop, and while loop) are indeed control structures in Java.*

■ CODING SECTION (2.0/5 - 40.0%)

■ Q1. Write a Java program that reads a map of key-value pairs from stdin and finds the value by its key. **Input Format:** - Line 1: The number of key-val...

Your Answer:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt(); // number of key-value pairs
        Map<String, String> map = new HashMap<>();

        // Read key-value pairs
        for (int i = 0; i < n; i++) {
            String key = sc.next();
            String value = sc.next();
            map.put(key, value);
        }

        // Key to search
        String searchKey = sc.next();

        // Check and print result
        if (map.containsKey(searchKey)) {
            System.out.println(map.get(searchKey));
        } else {
            System.out.println("Key not found");
        }

        sc.close();
    }
}
```

■ *Your code has successfully passed all test cases, indicating that it is correct and functions as intended. Great job on writing accurate and effective code to solve the problem!*

■ Q2. Write a Java program that reads an array from stdin and rotates it. **Input Format: - Line 1: The number of elements - Line 2: The array elements (...**

Your Answer:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt(); // number of elements
        int[] arr = new int[n];

        // read array elements
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        // store first element
        int first = arr[0];

        // shift elements left
        for (int i = 0; i < n - 1; i++) {
            arr[i] = arr[i + 1];
        }

        // place first element at the end
        arr[n - 1] = first;

        // print rotated array
```

```java
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }

        sc.close();
    }
}
```

■ *Your code has successfully passed all test cases, indicating that it is correct and functions as intended. Great job on achieving an "Accepted" status, keep up the good work!*

■ Q3. Write a Java program that reads a graph from stdin and finds the shortest path using the Floyd-Warshall algorithm. **Input Format:** - Line 1: The nu...

Your Answer:

```java
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt(); // number of nodes
        int e = sc.nextInt(); // number of edges

        int INF = 99999;
        int[][] dist = new int[n][n];

        // Initialize distance matrix
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i == j)
                    dist[i][j] = 0;
                else
                    dist[i][j] = INF;
            }
        }

        // Read edges (u v w)
        for (int i = 0; i < e; i++) {
            int u = sc.nextInt();
            int v = sc.nextInt();
            int w = sc.nextInt();
            dist[u][v] = w;
        }

        // Floyd-Warshall Algorithm
        for (int k = 0; k < n; k++) {
            for (int i = 0; i < n; i++) {
                for (int j = 0; j < n; j++) {
                    if (dist[i][k] + dist[k][j] < dist[i][j]) {
                        dist[i][j] = dist[i][k] + dist[k][j];
                    }
                }
            }
        }

        // Example: shortest path from node 0 to others
        System.out.print("[");
        for (int i = 0; i < n; i++) {
            System.out.print(dist[0][i]);
            if (i < n - 1) System.out.print(", ");
        }
```

```
        System.out.println("]");

        sc.close();
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int numNodes = scanner.nextInt();
        int numEdges = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over

        int[][] graph = new int[numNodes][numNodes];
        for (int i = 0; i < numNodes; i++) {
            for (int j = 0; j < numNodes; j++) {
                if (i == j) {
                    graph[i][j] = 0;
                } else {
                    graph[i][j] = Integer.MAX_VALUE;
                }
            }
        }

        for (int i = 0; i < numEdges; i++) {
            String[] edge = scanner.nextLine().split(" ");
            int source = Integer.parseInt(edge[0]);
            int destination = Integer.parseInt(edge[1]);
            int weight = Integer.parseInt(edge[2]);
            graph[source][destination] = weight;
        }

        // Floyd-Warshall algorithm
        for (int k = 0; k < numNodes; k++) {
            for (int i = 0; i < numNodes; i++) {
                for (int j = 0; j < numNodes; j++) {
                    if (graph[i][k] != Integer.MAX_VALUE && graph[k][j] != Integer.MAX_VALUE) {
                        int newDistance = graph[i][k] + graph[k][j];
                        if (newDistance < graph[i][j]) {
                            graph[i][j] = newDistance;
                        }
                    }
                }
            }
        }

        // Find shortest path
        int[] shortestPath = new int[numNodes];
        int currentNode = 0;
        for (int i = 0; i < numNodes; i++) {
            shortestPath[i] = currentNode;
            int nextNode = -1;
            int minDistance = Integer.MAX_VALUE;
            for (int j = 0; j < numNodes; j++) {
                if (graph[currentNode][j] < minDistance && graph[currentNode][j] != 0) {
                    minDistance = graph[currentNode][j];
                    nextNode = j;
                }
```

```
            }
            currentNode = nextNode;
        }

        System.out.print("[");
        for (int i = 0; i < numNodes; i++) {
            System.out.print(shortestPath[i]);
            if (i < numNodes - 1) {
                System.out.print(", ");
            }
        }
        System.out.println("]");

        scanner.close();
    }
}
```

■ *The output differs from the expected output because the actual output `[0, 2, 5, 6]` does not represent a valid path in the graph, whereas the expected output `[0, 1, 2, 3]` appears to be a correct sequence of nodes. The issue seems to be with the implementation of the Floyd-Warshall algorithm or the way the shortest path is being constructed from the distance matrix.*

■ Q4. Write a Java program that reads an array of integers from stdin and checks if the index is within bounds. **Input Format:** - Line 1: The number of e...

Your Answer:

```
public class Main {
    public static void main(String[] args) {
        // Write your solution here
        solution();
    }

    public static void solution() {
        // Your code here
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int n = scanner.nextInt();
        int[] array = new int[n];

        for (int i = 0; i < n; i++) {
            array[i] = scanner.nextInt();
        }

        int index = scanner.nextInt();

        if (index >= 0 && index < n) {
            System.out.println(array[index]);
        } else {
            System.out.println("ERROR: Index out of bounds");
        }

        scanner.close();
    }
}
```

■ *The code failed to produce any output, resulting in an empty string, whereas the expected output was '3'. This suggests that the code did not successfully read the array elements and access the element at the given index. The issue likely lies in the implementation of the solution method, which is currently empty.*

■ Q5. Write a Java program that reads an integer from stdin and checks if it's a valid integer. **Input Format:** - Line 1: A string **Output Format:** - ...

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
import java.util.InputMismatchException;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        try {
            int number = scanner.nextInt();
            System.out.println(number);
        } catch (InputMismatchException e) {
            System.out.println("Invalid integer");
        } finally {
            scanner.close();
        }
    }
}
```

■ *There is no test case failure provided to analyze. The student's code is also not submitted, so it's impossible to determine what went wrong. No feedback can be given based on the provided information.*

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.